

# EcoTrack Database Design

This document defines the initial database design for the EcoTrack MERN application. It serves as the agreed blueprint for backend development.

## Design Process

- Understand the project requirements.
- Identify the entities (things that need to be stored).
- Define the attributes (fields) for each entity.
- Define relationships between entities.
- Create MongoDB collections and Mongoose models.

## Entity Relationships

Company (1) → Many Users

Company (1) → Many Departments

Department (1) → Many Carbon Logs

User (1) → Many Carbon Logs (createdBy)

## Users Collection

| Field     | Purpose              |
|-----------|----------------------|
| _id       | MongoDB ObjectId     |
| name      | User full name       |
| email     | Unique email         |
| password  | Hashed password      |
| role      | Admin/User           |
| companyId | Reference to Company |
| createdAt | Timestamp            |
| updatedAt | Timestamp            |

## Companies Collection

| Field       | Purpose                  |
|-------------|--------------------------|
| _id         | MongoDB ObjectId         |
| companyName | Company name             |
| region      | Primary operating region |
| createdBy   | Reference to User        |
| createdAt   | Timestamp                |

## Departments Collection

| Field     | Purpose              |
|-----------|----------------------|
| _id       | MongoDB ObjectId     |
| name      | Department name      |
| companyId | Reference to Company |
| createdAt | Timestamp            |

## CarbonLogs Collection

| Field            | Purpose                      |
|------------------|------------------------------|
| _id              | MongoDB ObjectId             |
| companyId        | Reference to Company         |
| departmentId     | Reference to Department      |
| activityType     | Electricity/Fuel/Travel/etc. |
| amount           | Raw value                    |
| unit             | kWh, km, litres, etc.        |
| carbonEquivalent | Calculated CO2e              |
| date             | Activity date                |
| createdBy        | Reference to User            |
| notes            | Optional remarks             |
| createdAt        | Timestamp                    |
| updatedAt        | Timestamp                    |

## Recommended Activity Types

Electricity, Fuel, Business Travel, Air Travel, Road Transport, Waste, Water Usage, Manufacturing, Other.

## Notes

This design is intentionally simple for the summer-term project. Additional collections (e.g., Reports, Notifications, Audit Logs) can be introduced later if required.